

Response to Chin et al. 2026

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Introduction

In his commentary, Chin argues that our earlier paper is best understood as identifying methodological flaws in the firearms validity literature, not as showing that those flaws necessarily render the affected studies invalid (Chin 2026). On one reading, that is a fair caution. Not every flaw has the same consequences, and not every flaw defeats the possibility of learning something from a study. Some problems are primarily analytic. If a study reports error rates without confidence intervals, those intervals can often be computed later from the observed data. Likewise, if inconclusive responses were excluded from one reported summary, alternative summaries can often be calculated from the same response data. These are genuine shortcomings, but they are shortcomings that may be at least partially repaired after data collection.

Our disagreement with Chin is therefore narrower and more precise than a blanket claim that any flaw invalidates a study. The central point of our earlier paper was that some flaws are substantially more serious than others because they operate at the level of design and sampling rather than downstream analysis (Cuellar et al. 2024). When the sampled examiners are not representative of the population about which claims are being made, or when the sampled comparisons do not reflect the range and difficulty of items encountered in practice, the resulting data cannot support generalizable estimates of validity. In those cases, the problem is not simply that the wrong quantity was reported. The problem is that the data needed to estimate the relevant quantity were never collected in the first place.

That distinction matters because post hoc statistical corrections can only work with information that is already in the dataset. If the raw responses exist and the issue is only that the reported summary was incomplete or poorly chosen, then reanalysis may solve the problem. But if the study design failed to capture the relevant variation in examiner population, item difficulty, or testing conditions, no amount of reanalysis can recover what was never observed. One cannot reconstruct field-relevant error rates from a convenience sample of examiners, nor can one infer operational accuracy from a set of test items that systematically understates or distorts real-world difficulty. For these design-based flaws, the resulting estimates are not merely imperfect. They are not valid estimates of the performance quantity that courts, policymakers, and researchers care about.

This response therefore draws a distinction between remediable and non-remediable flaws. We agree with Chin that some defects in published firearms studies do not justify discarding those studies wholesale. But we maintain that the most serious flaws identified in our earlier paper are not minor imperfections awaiting statistical repair. They prevent the existing studies from answering the question they are routinely cited to answer: what are the error rates of firearm examiners under conditions relevant to practice? Our aim here is to make that distinction explicit and to explain why some flaws can be corrected after the fact, while others require new data collected under a valid design.

References

- Chin, Jason M. 2026. "Towards Cumulative Forensic Science: A Commentary on Methodological Problems in Every Black-Box Study of Forensic Firearm Comparisons." *Law, Probability and Risk* 25 (1): mgaf016. <https://doi.org/10.1093/lpr/mgaf016>.
- Cuellar, Maria, Susan Vanderplas, Amanda Luby, and Michael Rosenblum. 2024. "Methodological Problems in Every Black-Box Study of Forensic Firearm Comparisons." *Law, Probability and Risk* 23 (1): mgae015.

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